Lab Report: Comparing CNN architectures

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. All experiments were run on a single GPU with 16 GB of memory.

2 Results

Figure 1 shows the validation accuracy by dataset.

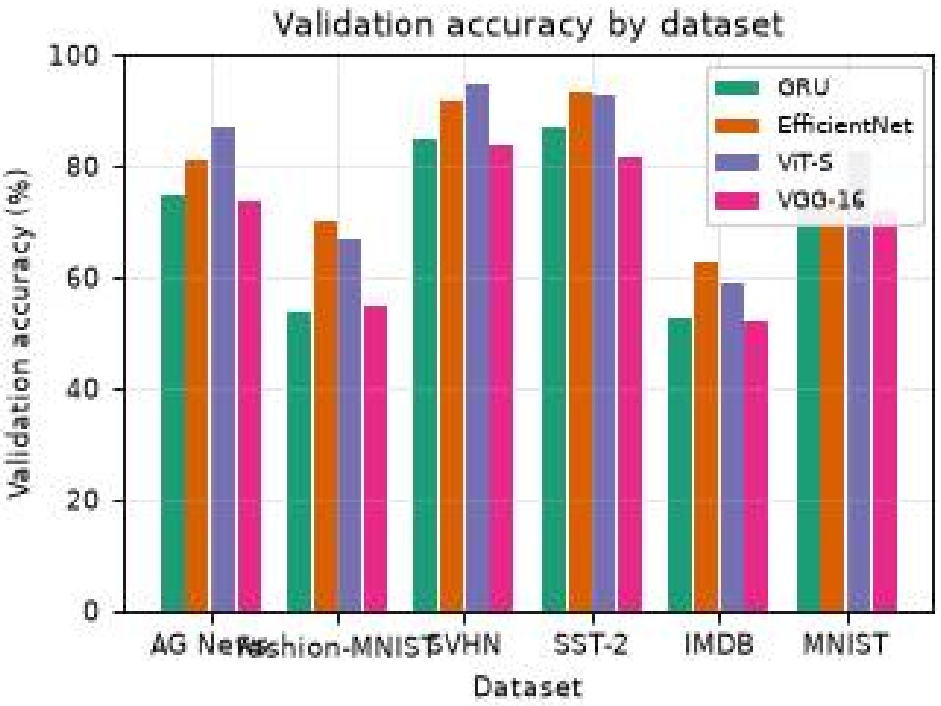


Figure 1: Validation accuracy by dataset for the models in this experiment.

As shown in Figure 1, ViT-S achieves 98.0% validation accuracy on SST-2. As shown in Figure 1, EfficientNet achieves 59.9% validation accuracy on Fashion-MNIST. ViT-S achieves 69.9% validation accuracy on Fashion-MNIST.

Figure 2 shows the validation accuracy during training.

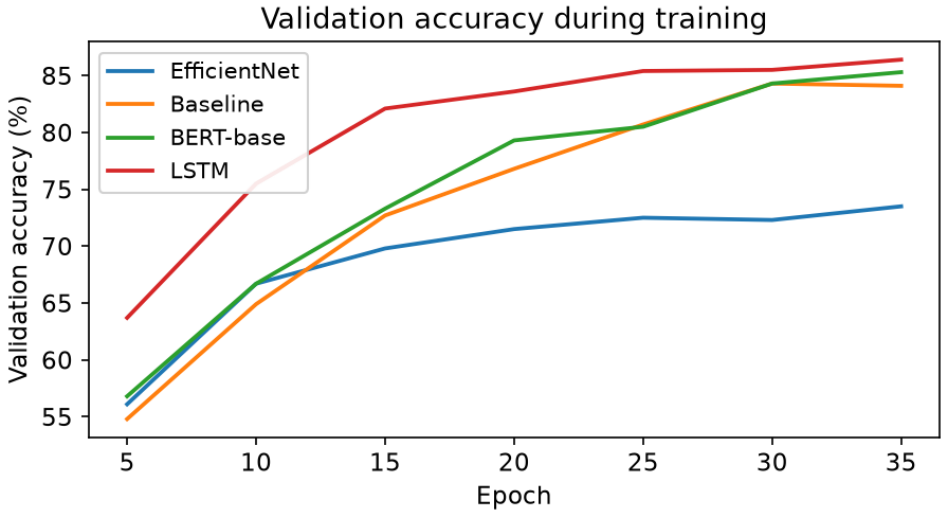


Figure 2: Validation accuracy during training for the models in this experiment.

As shown in Figure 2, LSTM outperforms Baseline by 1 percentage points at epoch 30. BERT-base attains a

validation accuracy of 82.8% at epoch 20. From epoch 30 to epoch 35, the validation accuracy of EfficientNet improves by a relative 25.7%.

3 Discussion

We note that the training was stable for all models.